
25. Private school choice in developing countries: experimental results from Delhi, India

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INTRODUCTION

India has the largest concentration of poor people in the world. About 35 percent of the country's population lives in poverty and over a third of the total population is illiterate, a figure that is closer to 49 percent for females (Central Intelligence Agency, 2006). While government schools for the poor struggle to improve educational outcomes such as basic literacy and numeracy skills, low-cost private alternatives can be found in both urban and rural areas. This private school choice study focuses on a 20 square kilometer, highly urbanized slum area known as Shahdara, situated in East Delhi on the banks of the Yamuna River. As is the case across much of the developing world (e.g. Dixon, 2013; Tooley, 2009), a low-cost private education market targeted towards economically disadvantaged families has emerged in Shahdara (Aggarwal, 2000; Alderman et al., 2001, 2003; Baurer et al., 2002; De et al., 2002; Drèze and Sen, 2002; Nambissan, 2003; Rose, 2002, 2003; Watkins, 2000). Only a little over one-fourth of all schools in Shahdara are government schools and the rest are private (Tooley and Dixon, 2007). It is popular to claim that such low-cost private schools must be inferior to the higher-cost government options (Watkins, 2000), yet little empirical evidence exists to confirm or deny this claim.

Private school vouchers are subsidies provided to parents to enable them to enroll their child in a private school of their choosing so long as the school agrees to accept the voucher as full or partial payment for educating the child. Generally they are referred to as "vouchers" if they are provided to families by governments and "scholarships" if the funding comes from charitable sources. For the purposes of this chapter, we will refer to any arrangement whereby resources are provided to families to allow them to enroll in a private school as a school voucher, regardless of the source of the funds.

Few topics in education policy are more contentious than the

desirability and effects of school vouchers. Supporters argue that expanding school choice through vouchers will improve student outcomes in three ways: (1) by facilitating access to private schools that are better-managed than government schools; (2) by providing parents with more educational options so that they can better match their child to the school that best fits the child's needs; and (3) by generating market-based competitive pressure for all schools to improve or risk losing customers and thus resources (Friedman and Friedman, 1981; Chubb and Moe, 1990; Hoxby, 2003). Supporters also claim that access to private schools for disadvantaged students, through vouchers, will reduce educational inequalities because advantaged students have access to the best government-run and privately run schools under the status quo (Brighouse, 2000; Viteritti, 1999). Voucher opponents counter that government-run schools are as good as private schools, once one accounts for differences in the populations they serve, and that private school vouchers will exacerbate inequalities because mostly advantaged students will use them (Henig, 1999; Elmore and Fuller, 1996; Berliner and Biddle, 1995). Centrist researchers have argued that either side of the debate might be correct, depending on the local context and policy design of school vouchers (National Working Commission on Choice in K-12 Education, 2003; Hess, 2002).

There are three primary reasons why Shahdara parents, in particular, may seek and benefit from a private school alternative to the free, government schools that are available. The first two reasons are teacher absenteeism and low levels of teacher effort in government schools, which have led many parents across the developing world to consider private schooling options (Muralidharan and Sundararaman, 2013; Lewis and Patrinos, 2011). Comparing private and government schools in Shahdara, for example, Tooley and Dixon (2007) report that private school teachers were more likely than their government school counterparts to be teaching when researchers visited the school unannounced. The third reason why Indian parents may want their children to attend a private school is because such schools are more likely to teach all subjects in English, which is not the case in the government schools (Tooley and Dixon, 2002). Although Hindi is the most widely spoken language in India, English is the most important language for communication in politics and commerce and is used everywhere as a bridge between the many official languages and dialects (Mitra et al., 2003). As such, parents in India may reasonably expect that fluency in English will improve their children's opportunities in the job market and in higher education.

The ENABLE Program

For many Indian parents, access to a private school may be out of reach because of the expenses involved and insufficient information about available private school options. In response to this need, the UK-based charity, Absolute Return for Kids (ARK), partnered with the Centre for Civil Society in 2011 to create an education voucher program called Ensure Access to Better Learning Experiences (ENABLE). The ENABLE program is targeted to economically disadvantaged children and is designed to encourage families to claim free private school places, guaranteed to them under the Right of Children to Free and Compulsory Education Act of 2009, which established the right of every child aged 6–14 to a free, full time elementary education. Many Indian parents are not aware of this right or may struggle with the associated expenses (e.g. uniforms, textbooks). To date, the ENABLE program has granted vouchers worth up to 7,300 rupees annually (about \$117 US as of February 2015) to over 800 pupils to attend a low-cost private school. To be eligible for a voucher, family income could not exceed Rs. 8,000 per month (around \$129 US) and the child had to be attending a government school, attending an unrecognized private school, or not attending school at all. Approximately 1,800 children applied for vouchers and 100 private schools agreed to accept voucher students, from which parents could choose their top three schools.

ENABLE vouchers were allocated via lottery, allowing researchers to structure an impact evaluation as a randomized controlled trial, also known as a field experiment. Students who won the voucher lottery are compared to students who lost the lottery, with the only differences between the two groups being the award of voucher and random chance. Field experiments often are referred to as the “gold standard” in research design, since they generate unbiased estimates of the causal relationship between a given intervention and subsequent outcomes (Wolf et al., 2013, p.249; Tufte, 2006, p. 145; Boruch et al., 2002, p.740). Those students in the experimental treatment group actually receive four vouchers in all, covering tuition, uniforms, textbooks, and lunch for the full five years of primary school. ENABLE also provides a handbook to parents to help them identify private schools, judge school quality, and choose the most appropriate school for their child.

PREVIOUS STUDIES OF SIMILAR PROGRAMS

Much is known with confidence about the participant effects of school vouchers in the United States. Specifically, there have been ten

Table 25.1 Achievement impacts of school vouchers in the United States, peer-reviewed, random assignment studies

Author(s)	Year	Location	Findings
Wolf, Gutmann, Puma, Kisida, Rizzo, and Eissa	2013	DC	Positive Overall
Cowen	2008	Charlotte	Positive Overall
Howell, Wolf, Campbell, and Peterson	2002	DC	Positive Overall
Greene	2001	Charlotte	Positive Overall
Greene, Peterson, and Du	1998	Milwaukee	Positive Overall
Rouse	1998	Milwaukee	Positive Overall
Howell, Wolf, Campbell, and Peterson	2002	Dayton, OH	Positive for Subgroups
Jin, Barnard, and Rubin	2010	New York	Positive for Subgroups
Barnard, Frangakis, Hill, and Rubin	2003	New York	Positive for Subgroups
Howell, Wolf, Campbell, and Peterson	2002	New York	Positive for Subgroups
Mayer et al.	2002	New York	Positive for Subgroups
Krueger and Zhu	2004	New York	Null Results

peer-reviewed, random-assignment studies of school voucher programs that present 12 analyses of their achievement effects on participants (Table 25.1). Six analyses find positive results overall: Wolf et al. (2013); Cowen (2008); Howell et al. (2002, DC Results); Greene (2001); Greene et al. (1998); and Rouse (1998). Furthermore, an additional five analyses find significant benefits for at least one important student subgroup, such as African American students (Howell et al., 2002, New York and Dayton results; Jin et al., 2010; Barnard et al., 2003; Mayer et al., 2002). Only one random-assignment study of school voucher impacts reports no significant effects on test score outcomes (Krueger and Zhu, 2004).

A notable weakness of this body of evidence is that most of the experimental studies of private school vouchers have been conducted in the United States. We know of only three random assignment studies of the impacts of school vouchers on student achievement that involve programs outside of the United States. One study is of a partial-tuition school voucher program in Colombia (Angrist et al., 2002). The researchers describe it as a “natural experiment” instead of a field experiment because they did not design the voucher lottery for research purposes but simply benefited from the fact that the government had distributed the vouchers by lottery as a matter of fairness. They report that students scored significantly higher on achievement tests three years later if they had been awarded a voucher. The positive impact was especially large for girls.

Two experimental studies of school vouchers have been set in India but involve non-ENABLE private school voucher programs. One study

is of a school voucher intervention in the Indian state of Andhra Pradesh (Muralidharan and Sundararaman, 2013). The researchers report that the opportunity to attend a private school of choice through the voucher resulted in students scoring the same as the control group in the areas of mathematics and their native language (Telugu) but significantly higher across the disciplines of English, science, social studies, and Hindi. The second study is of a voucher program in the region of North East Delhi that exclusively serves girls (Centre for Civil Society, 2014). The researchers find that the students awarded vouchers scored significantly higher than the randomized control group students in English, mathematics, and Hindi after four years, with the largest score increase of 10 percent in English.

In sum, previous rigorous experimental evaluations of private school vouchers have found that students who receive access to schools of choice through such programs tend to perform at least as well as, and often better than, their control group peers on subsequent achievement tests. Although most of the experimental research on vouchers is limited to the United States, two random assignment studies of partial-tuition vouchers in regions of India have reported positive achievement effects, with the gains especially large for girls and in English. Two rigorous studies of the impacts of school vouchers in a large country like India, however, represent a thin reed on which to rest policy recommendations. Our study seeks to complement and extend this early research on private school vouchers in India by providing results from a third rigorous evaluation of a private school voucher program.

METHODOLOGY

Before the lottery was conducted, all applicants were visited in their homes, where they took baseline achievement tests in Hindi, mathematics (Part A and B), and English. Families also completed background surveys that collected information on demographic characteristics, parental educational levels, employment, family assets, and expectations regarding the voucher program. At the end of one and two years in the voucher program, students were tested again in the three subjects: Hindi, mathematics, and English. Changes in achievement between these two time periods generate the impact estimates of the ENABLE program.

Three methodological considerations are central to our analysis. First, although the random assignment of students through the lottery means that the treatment and control groups are similar in expectation regarding all student factors, both measurable and unmeasurable, we include

baseline covariates in the calculation of the impact estimate to make the estimation more precise and therefore maximize the analytic power of the evaluation.

Second, we use weighting adjustments to compensate for non-response in test score data in the three subjects, Hindi, English, and mathematics. Non-response could lead to biased estimates if the mechanism by which certain individuals provide outcome data while others do not is related to the outcome being examined (Groves et al., 2004). These adjustments help protect against potential biases by ensuring that the weighted distribution of outcome data from respondents is similar to the original study population as a whole on the basis of all observed variables.

The third methodological decision relates to test sensitivity. The ability of the evaluation to identify the impact of the program on academic outcomes is largely the function of two factors: the magnitude of the true program impact and the sensitivity of the test in pin-pointing the actual achievement level of the students, especially at outcome but also at baseline. The consideration of test sensitivity is especially important when testing young children from disadvantaged backgrounds. Tests that produce many scores of “0” or “100” tend to be highly insensitive because such “floors” and “ceilings” mask the actual achievement level of the student and therefore render it impossible to estimate changes accurately. The more advanced B section of the mathematics test generated zero scores for 88 percent of the students at baseline and zeroes at both baseline and in year two for 15 percent of the sample. On the other hand, the easier A section of the mathematics test produced a large number of non-zero scores. In the results below, therefore, we present the impacts of the program on mathematics scores only for part A, since that test is the gauge of mathematics performance most suitable for discerning changes in mathematics achievement over time.

EFFECTS OF ENABLE ON STUDENT ACHIEVEMENT

Impact of Voucher Offer

Table 25.2 presents the two-year impact estimates of the offer of a private school voucher, commonly referred to as Intent to Treat (ITT) estimates. The primary analysis revealed a positive and statistically significant impact of the voucher program on English language achievement. Students in the treatment group scored, on average, 2.23 points higher on the outcome English assessment. This is equal to 0.15 of a standard deviation (SD) or an advantage of 22.6 percent in English achievement for the voucher

Table 25.2 Two-year impact estimates of the offer of a voucher on academic achievement, all subjects

Subject	Impact of the voucher offer (ITT)				
	Treatment group mean	Control group mean	Difference (estimated impact)	Effect size	p-value of estimate
English	12.11	9.88	2.23***	0.15	0.00
Mathematics	15.56	15.18	0.38	0.07	0.21
Hindi	13.98	13.13	0.85	0.04	0.39

Notes: Means are regression adjusted using a consistent set of baseline covariates. Impacts are displayed in terms of scale scores. Effect sizes are in terms of standard deviations. Separate mathematics, English, and Hindi response weights used. Impacts represent ITT estimates and thus have not been adjusted for rates of usage; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

treatment group relative to the control group. The positive impact of the program on student English achievement was statistically significant beyond the 99 percent confidence level, which is the highest level used in the analysis. Although the treatment group on average scored higher than the control group on Hindi language and mathematics as well, those differences were not statistically significant and therefore may be a product of random error.

Baseline covariates included in the model reveal relationships that are all in the expected direction – baseline mathematics, English, and Hindi scores, the child's age, and average monthly income all have predictably positive effects on achievement. Similarly, we find predictably negative achievement effects associated with having higher numbers of children in the house, illiterate parents/guardians, and use of a community toilet as compared to having a toilet in the family's house or having access to a shared toilet. The fact that these measures of crucial child and family resources influence student English outcome scores in predictable ways lends added credibility to the results of the analysis.

The absence of a clear effect of the ENABLE program on Hindi achievement could be due to students in both the treatment and control groups similarly obtaining much of their Hindi language skills at home or in their neighborhoods. In contrast, skills in a second language such as English, as well as mathematics, are more school-dependent and therefore better indicators of the benefits of a school choice intervention such as ENABLE. The impact of the program on English achievement is positive, clear, and moderate in size. The impact of the program on

mathematics, though positive, is uncertain, given an effect estimate very close to zero.

It is important to note that the lottery losers who accessed private schooling using their own resources, comprising about one-fourth of all control group students, remained in the study as control group students. This action is necessary to preserve the purity of the random assignment of students and is a central feature of ITT estimates in field experiments. Any gains that control group students realized from experiencing private schooling themselves were averaged in with the results from control group students who stayed in public schools. Thus, the analyses of the pure experimental effect of being offered an ENABLE voucher presented here are conservative because some students offered vouchers didn't use them (11 percent of students with second year outcome data) and a substantial portion of the control group joined the voucher users in private schools (31 percent of students with second year outcome data). To correct for this, we present the results of an instrumental variables analysis in the next section to generate unbiased estimates of the effect of private schooling that account for both treatment students who do not attend private schools and control group students who do.

Impact of Voucher Use

Using the voucher lottery as an instrumental variable recovers unbiased estimates of the effect of actually using a voucher to attend a private school in the context of a voucher experiment that includes treatment decliners and control crossovers (Howell et al., 2006, pp.49–51). Since many policy makers are most interested in the effect of actually experiencing private schooling as opposed to government schooling, and not the pure experimental impact of being offered a voucher, we present the instrumental variables analysis results as our estimate of the impact of the treatment of private schooling on the treated, that is the students who actually used a voucher.

Applying instrumental variable analytic methods to the ENABLE data, the positive impacts on student English outcomes remain statistically significant (Table 25.3). The effect of attending private school on student English achievement, based on the patterns in the data, is an average increase of 3.65 points or 0.25 of a standard deviation and is statistically significant beyond the 99 percent confidence level. The effects of private schooling on mathematics and Hindi achievement are also positive but not significantly different from zero.

Table 25.3 Two-year impact estimates of the use of a voucher on academic achievement, all subjects

Subject	Impact of scholarship use (IOT)		
	Estimated impact	Effect size	<i>p</i> -value of estimate
English	3.65***	0.25	0.00
Mathematics	0.62	0.12	0.20
Hindi	1.39	0.07	0.38

Notes: Impacts are displayed in terms of scale scores. Effect sizes are in terms of standard deviations. Regression weights are used to adjust for non-response. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Subgroup Analyses

To investigate whether subgroups respond to the intervention differently, we examine achievement impacts by gender. The education of girls is a particular concern in India, as some families in poverty concentrate their resources on their male children. In our analysis we see a consistent pattern across the three domains of English, mathematics, and Hindi, with female students demonstrating a clearer positive impact from the program than male students.

In the left panel of Table 25.4 we start with the pure experimental ITT estimates. They reveal that the impact of the program in English for girls was, itself, statistically significant and was significantly higher than for boys. That is, the ENABLE program clearly increased the English achievement of the subgroup of girls in the study, and the positive impact of the program on English achievement was significantly greater for girls than for boys. The difference in the estimated impact between males and females is 2.44 points, which is statistically significant with at least 90 percent confidence. Similarly, we observe a positive impact of the program in mathematics on the subgroup of female students. We can be at least 90 percent confident that the impact of the program on mathematics achievement differed by gender, with girls gaining nearly 1 scale score point if given a voucher while the program had no discernable impact on the mathematics scores of boys. Finally, we observe that the impact of the program on Hindi language skills did not differ significantly between boys and girls, though the subgroup of girls in the study scored significantly higher in Hindi if they were offered a private school voucher.

The right panel of Table 25.4 displays the Impact on Treated (IOT) estimates associated with voucher use, again using the lottery as an

Table 25.4 *Impact estimates of the offer and use of a voucher on subgroups two years after application*

	Impact of voucher offer (ITT)			Impact of voucher use (IOT)		
	Male	Female	Difference	Male	Female	Difference
ENGLISH						
Treatment group mean	11.81	12.45	-0.64	12.49	14.73	-2.24
Control group mean	10.72	8.92	1.80	10.72	8.92	1.80
Difference (estimated impact)	1.09	3.52***	-2.44*	1.77	5.81***	-4.04*
Effect size	0.07	0.27	-0.20	0.11	0.44	-0.33
<i>p</i> -value of estimates	0.27	0.00	0.09	0.26	0.00	0.08
MATHEMATICS						
Treatment group mean	15.28	15.87	-0.59	15.25	16.44	-1.19
Control group mean	15.35	15.00	0.35	15.35	15.00	0.35
Difference (estimated impact)	-0.06	0.88**	-0.94*	-0.10	1.44**	-1.55*
Effect size	-0.01	0.18	-0.19	-0.02	0.29	-0.31
<i>p</i> -value of estimates	0.87	0.04	0.11	0.88	0.04	0.11
HINDI						
Treatment group mean	13.12	14.94	-1.81	12.80	16.49	-3.69
Control group mean	13.66	12.52	1.13	13.66	12.52	1.13
Difference (estimated impact)	-0.53	2.41*	-2.95	-0.86	3.97*	-4.82*
Effect size	-0.03	0.15	-0.17	-0.04	0.24	-0.28
<i>p</i> -value of estimate	0.69	0.09	0.13	0.69	0.09	0.13

Notes: Means are regression adjusted using a consistent set of baseline covariates. Impacts are displayed in terms of scale scores. Effect sizes are in terms of standard deviations. Response weights used. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

instrumental variable to predict the achievement of students who actually use vouchers to attend private schools through the ENABLE program. In all three subjects, we observe that the impact of actually using a private school voucher on females is to increase their achievement by a statistically significant amount, and those program benefits are larger for girls than for boys. The difference in the estimated impact of the program between males and females in English is 4.04 points. In mathematics, the difference is 1.55 points and in Hindi the difference is 4.82 points. All three estimates are statistically significant at $p < 0.10$ so we can be at least 90 percent confident that the impact of voucher usage differed by gender.

ASSESSING THE MAGNITUDE OF THE ENABLE PROGRAM'S IMPACTS

The statistical analysis revealed a positive and statistically significant impact of the ENABLE program on student achievement in English as well as statistically significant impacts specifically for girls in all three subject domains of English, mathematics, and Hindi. How big were the gains for the voucher students? The average gain of 0.15 SD in English achievement across the entire group of scholarship winners was equivalent to a gain of 4.5 months of additional learning in English in the 18 month learning period between the start of school and the administration of the second year outcome tests. The statistically significant achievement gains from winning the ENABLE lottery specifically for girls amounted to nearly 8 additional months of learning in English, about 3 additional months of learning in mathematics, and 4 additional months of learning in Hindi.

The magnitude of the English gains from the ENABLE program administered by ARK also can be gauged by comparing them to the two-year experimental impacts from other prominent education interventions (Table 25.5; Figure 25.1). The DC school voucher evaluation (DC Choice), the rigorous teacher merit pay study, the large education technology evaluation, and the famed STAR class size reduction study all reported second year experimental impact estimates that were much smaller than the ARK English gains.

CONCLUSION

This evaluation of ARK's ENABLE school voucher program in Shahdara, Delhi is one of only a select few experimental studies of such programs outside of the United States. Because students were randomly assigned to treatment and control groups at baseline, we can be certain that any differences in achievement outcomes across these groups reflect the true program impact and are not the effect of any observable or unobservable student characteristics. All students were tested in the areas of English language, Hindi language, and mathematics at baseline and again after two years in the program. The results are generally consistent with previous evidence on this educational intervention, especially the two previous experimental studies of private school vouchers in India. While there were no observable differences in Hindi or mathematics, the treatment group clearly outperformed the control group in English, with those students who were offered a voucher scoring 2.23 points higher on the English

Table 25.5 Comparing ARK results to other high profile educational interventions

Intervention	Authors	2-year effect size	Significant?
Education Technology – Impact on Reading	Campuzano et al. (2009)	–0.06	NO
Teacher Induction – Impact on Mathematics	Isenberg et al. (2009)	–0.02	NO
Teacher Induction – Impact on Reading	Isenberg et al. (2009)	0.00	NO
DC Choice Mathematics Achievement	Wolf et al. (2008)	0.01	NO
STAR Class size reduction – Impact on Reading (1st Grade)	Ding and Lehrer (2010)	0.02	NO
STAR Class size reduction – Impact on Mathematics (1st Grade)	Ding and Lehrer (2010)	0.02	NO
Teacher Merit Pay– Impact on Mathematics	Springer et al. (2010)	0.04	NO
DC Choice Reading Achievement	Wolf et al. (2008)	0.09	NO
ARK – Impact on English	Wolf et al. (2013)	0.15	YES

Note: Effect sizes are in standard deviation units and represent impacts on student achievement after two years.

exam as a result of being offered an ENABLE voucher, equivalent to a gain of 4.5 months of additional learning in English.

Subgroup analyses reveal that the ENABLE program was particularly beneficial for girls. Instrumental variable estimates of actually using a voucher reveal a statistically significant difference in English, mathematics, and Hindi by gender, with females consistently benefiting more from the intervention than their male counterparts. In fact, the ENABLE program effectively reversed the typical pattern of gender bias in education in India. For the control group, the subgroup of boys in the study consistently outperformed the subgroup of girls on the three achievement tests. For the treatment group, however, the positive impact of the ENABLE program was so strong that they not only outperformed the control group girls on all three tests, the treated girls also outscored their male counterparts in the treatment group. At this early point we do not know why the ENABLE program was especially effective in increasing the academic achievement of girls, but it is encouraging that girls benefited more than boys given the unfortunate tradition of gender discrimination

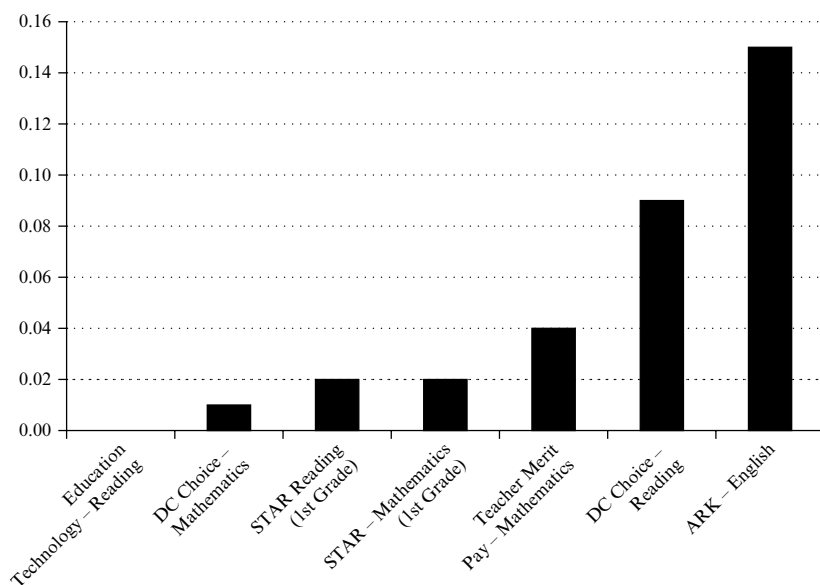


Figure 25.1 *Experimental achievement impacts from high profile education interventions*

against girls in the country. Most importantly, this rigorous experimental evaluation of the ENABLE voucher program provides causal evidence of the positive impact of a targeted school voucher program on the most important student outcome, English language achievement, for students in a slum area of East Delhi, India.

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